No. of Printed Pages : 4

MCS-226

MASTER OF COMPUTER APPLICATIONS (MCA) (NEW)

Term-End Examination

June, 2022

MCS-226 : DATA SCIENCE AND BIG DATA

Time : 3 Hours *Maximum Marks : 100*

(Weightage : 70%)

Note : (i) Question No. 1 is compulsory and carries 40 marks.

(ii) Attempt any ***three*** *questions from the remaining Q. Nos. 2 to 5.*

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1. (a) What is the Data Science ? What are its applications ? Explain the terms categorical and quantitative data with the help of an example for each. 5

(b) Assume that a fair coin is tossed 4 times. What would be the probability distribution of a variable X, which represents the number of tails in four tosses of coin ? What kind of probability distribution is this ? Justify. 5

(c) What is Big data ? Explain the characteristics of Big data. 5

(d) What is the map-reduce paradigm ? Explain with the help of an example. 5

(e) Define the term similarity. Explain the Jaccard similarity of sets with the help of an example. 5

(f) What is a data stream ? How is data stream processing different from transaction processing ? 5

(g) Write a program using R to add and subtract two matrices. 5

(h) Define the term classification. Write the steps about how R programming can be used to perform classification. 5

2. (a) What are the different measures for defining the spread of a quantitative variable ? Explain. 5

(b) What is meant by 'Sampling Distribution' ? Explain with the help of an example. 5

(c) Explain the concept of data pre-processing and data cleaning with the help of an example for each. 5

(d) What is a box plot ? Draw a sample box plot and explain it. 5

3. (a) Explain the characteristics of HDFS. What are the advantages of HDFS over file allocation table (FAT) based system ? 5

(b) What is the role of shuffling and sorting in map-reduce ? Explain with the help of an example. 5

(c) What are the features of SPARK architecture ? Explain. 5

(d) What are NoSQL databases ? How are they different from relational database system ? 5

4. (a) What is collaborative filtering ? Explain with the help of an example. 5

(b) What is the use of Bloom filtering ? Explain with the help of an example. 5

(c) Explain any page ranking algorithm with the help of an example. 5

(d) Explain the issues related to mining of social networks. 5

5. (a) Define the following in the context of R programming, with the help of an example : 5

(i) Complex data type

(ii) % % operator

(iii) ← or << – operator

(iv) Vectors

(v) Strings

(b) What is the use of bar chart ? How can you draw a bar chart using R programming language ? Explain. 5

(c) What is linear regression ? Explain how R programming can be used to perform linear regression. 5

(d) What is clustering ? Write the steps of how R-programming can be used to perform K-mean clustering. What is confusion matrix in this context ? 5

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MCS-226

MASTER OF COMPUTER APPLICATIONS (MCA-NEW)

Term-End Examination

December, 2022

MCS-226 : DATA SCIENCE AND BIG DATA

Time : 3 hours | *Maximum Marks : 100*
Weightage : 70%

Note : *Question no.* ***1*** *is* ***compulsory*** *and carries 40 marks. Attempt any* ***three*** *questions from the rest.*

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1. (a) Explain the following types of data : *6*

(i) Semi-structured data

(ii) Unstructured data

(iii) Qualitative data

(iv) Quantitative data

(b) What is meant by "Probability distribution of continuous random variable" ? Explain with the help of a diagram. Also explain the normal distribution. *6*

(c) What are the characteristics of Hadoop Distributed File System (HDFS) ? Why is it used for Big data processing ? *6*

(d) Explain the characteristics of data streams. *4*

(e) What are NoSQL databases ? Why are they used ? *4*

(f) Explain any one mechanism of filtering of data streams. *4*

(g) Explain the following, with the help of an example, in the context of R programming : *6*

(i) Dataframe

(ii) List

(iii) Vector

(h) What is logistic regression ? Which function of R programming can be used to implement logistic regression ? *4*

2. (a) Explain the characteristics of measurement scales of data. Use these characteristics to define various measurement scales of data. *6*

(b) Explain the steps of significance testing, with the help of an example. *8*

(c) Explain the following terms with the help of an example : *6*

(i) Data pre-processing

(ii) Data curation

(iii) Data cleaning

3. (a) Explain the characteristics of Big data. How does Big data differ from relational data ? *6*

(b) Explain the steps of map-reduce paradigm using the example of word counting. *6*

(c) List the features of any ***two*** of the following : *8*

(i) Apache Spark

(ii) Hive

(iii) Column-based databases

(iv) Graph-based databases

4. (a) How can link analysis be used to compute PageRank ? *4*

(b) Explain the concept of Recommendation System. *6*

(c) Explain how the similarity between two documents can be found. *6*

(d) Explain how the social networks can be represented using a graph. *4*

5\. (a) Write an R program to create two 3×3 matrices and multiply them. How is this program different from a similar C program ? *5*

(b) What is a box plot ? List the commands of R programming that can be used to create a box plot. *5*

(c) What is multiple regression ? Write steps about how R programming can be used to create multiple regression model. *5*

(d) What is a decision tree ? Write steps on how R programming can be used for making decision tree. *5*

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MASTER OF COMPUTER APPLICATIONS (MCA-NEW)

Term-End Examination

June, 2023

MSC-226 : DATA SCIENCE AND BIG DATA

Time : 3 Hours *Maximum Marks : 100*

Weightage : 70%

Note : *Question No. 1 is compulsory. Attempt any* ***three*** *questions from the rest.*

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1. (a) Define Data Science. Give advantages of Data Science in an organization. 5

   (b) Explain Bayes' Theorem with suitable equation and example. 5

   (c) What is a Histogram ? How does Histogram differ from Bargraph ? Briefly discuss the utility of Histogram in Data Science. 5

   (d) What is Hadoop MapReduce ? Give its advantages. Also, discuss how <key-value> pair mechanism facilitates MapReduce programming. 5

(e) In context of Data Science, what is Apache SPARK ? How does Apache SPARK differ from Hadoop ? 5

(f) What are Data Streams ? How do Data Streams differ from Databases ? Why mining of data streams is considered as a challenging process in Data Science ? 5

(g) Explain PageRank algorithm, with suitable example. 5

(h) What are Dataframes in 'R' programming ? Give characteristics of Dataframes. 5

2. (a) Write the syntax to create the following plots in 'R' : 5

(i) Bar charts

(ii) Box plots

(iii) Histogram

(iv) Line graphs

(v) Scatter plots

(b) Differentiate between Linear Regression and Multiple Regression, with suitable example for each. 5

(c) What are Decision Trees ? What are categorical variables and continuous variables ? How do these two variables relate to decision trees ? Explain the role of entropy and information gain in decision trees. 10

3. (a) Compare qualitative data with quantitative data. What do you understand by the term "Measurement scale of data" ? Give characteristics of measurement scales of data. List various measurement scales with suitable example for each. 10

(b) What is a Random Variable ? Differentiate between Discrete Random Variable and Continuous Random Variable. 5

(c) What is a Heat Map ? Give uses and best practices for Heat Maps. 5

4. (a) Explain the following operations of map-reduce with suitable example and supporting block diagram : 10

   (i) Splitting

   (ii) Mapping

   (iii) Shuffling

   (iv) Reducing

   (b) Explain the term Data lake. Briefly discuss the key capabilities of data lake. 5

   (c) Compare Land Mark Model and Sliding Windows Model for data stream processing. 5

5. Write short notes on the following : 4×5=20

   (a) Different mechanisms of finding PageRank

   (b) Chi-square test

   (c) Association Rules

   (d) Predictive Analysis

   (e) HIVE and its utility in Data Science